MAX PLANCK INSTITUTE
FOR EMPIRICAL AESTHETICS

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Postdoctoral Researcher

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Research interests

- Computational models of musical information for predictive modelling of neural and behavioural responses of affective experience
- Computational and experimental methodologies to understand structural and functional organisations of human brains

Positions

2021–Present Postdoctoral Researcher, Max Planck Institute for Empirical Aesthetics, Frankfurt am Main, Germany
- Supervisor: Dr. Daniela Sammler
- Projects: fMRI encoding of musical emotions and affective prosody

2018–2021 Postdoctoral research associate, Department of Psychology and Neuroscience, Duke University, Durham, NC, USA
- Supervisor: Dr. Tobias Overath
- Projects: MEG correlates of auditory percepts, fMRI encoding of natural speech and music

2017–2018 Postdoctoral research associate, Department of Psychiatry, Behavioural and Clinical Neuroscience Institute, University of Cambridge, Cambridge, UK
- Supervisor: Dr. Valerie Voon
- Projects: intracortical myelination and its association with impulsivity and alcohol consumption

Education and academic training

2012–2017 Doctor of natural science in psychology (Doctor rerum naturalium in Psychologie), International Max Planck Research School on Neuroscience of Communication (IMPRS-NeuroCom), Leipzig, Germany
- Advisors: Dr. Thomas R. Knösche and Dr. Erich Schröger
- Dissertation: “Myeloarchitecture and Intrinsic Functional Connectivity of Auditory Cortex in Musicians with Absolute Pitch” (*Summa cum laude*)

2010–2012 Research assistant, Department of Brain and Cognitive Sciences, Seoul National University, Seoul, South Korea
- Supervisors: Dr. Moo K. Chung and Dr. Jun Soo Kwon
- Projects: Graph theoretical analysis on structural covariance in clinical populations, subcortical structure shape analysis

2008–2010 Master of science in cognitive science, Seoul National University, South Korea

[1] My two-syllable first name reads [IPA: /sɯŋgu/]. Pragmatically, “SG” is also fine. :)

- Advisor: Dr. C. K. Chung
- Thesis: “The effect of conditional probability of chord progression in Western music corpus on brain response: an MEG study”

2000–2004 Bachelor of arts in economics and Bachelor of arts in psychology (dual majors), Yonsei University, South Korea

1996–1999 Specialised programme in musical composition, Sun Hwa Arts High School, South Korea

Military service

2004–2007 Military officer, Republic of Korea Air Force (mandatory service, honourably discharged as a first lieutenant)

Grants and funding

2026 Initiation of International Collaboration, German Research Foundation (DFG) and National Research Foundation of Korea (NRF)
- Role: Germany-side PI
- Funding: €7,600 (German side, DFG; 1 year)
- Bilateral workshop: “Affective Voice Synthesis: Industrial and Scientific Perspectives on Technology” (MPIEA and KAIST)

2026 R3 Career Support (funding for conference speakers), Johanna Quandt Young Academy, Goethe University Frankfurt, Frankfurt am Main, Germany
- Role: co-applicant
- Conference: CoBIC Opening Symposium, 20–21 April 2026

2025–2026 MPG Diversity Excellence Fund, Max Planck Society, Munich, Germany
- Role: PI
- Funding: €20,000 (1 year)
- Project: “EXPERIENCE — EXperimental PERformance series for Intercultural ENcounters & Collaborative Experience”

2023–2026 Academy Fellowship, Johanna Quandt Young Academy, Goethe University Frankfurt, Frankfurt am Main, Germany
- Role: PI
- Funding: €10,000 (2 years)
- Project: “Neural representation of affective experiences evoked by music”

2021 Postdoc Research Grant Award, The Charles Lafitte Foundation Program for Research in Psychology & Neuroscience, Duke University, NC, USA
- Project: “Temporal Processing in Tonal and Atonal Music” (Amazon MTurk)

Honours and awards

2017 Summa cum laude, PhD examination, Institute of Psychology, University of Leipzig, Germany

2014 Conference Scholarship, The 5th Conference for Neurosciences and Music, Mariani Foundation, Italy

2013–2016 Conference Scholarship, Research Academy Leipzig, University of Leipzig, Leipzig, Germany

2012–2016 Ph.D. Scholarship, International Max Planck Research School (IMPRS) on Neuroscience of Communication, Germany

2001–2004 B.A. Scholarship, Suam scholarship foundation, South Korea

Publications

*Auditory perception (*N = *11)*

2026 Baek S-C, **Kim S-G**, Maess B, Grigutsch M, Sammler D. Dynamic acoustic-to-categorical representations of phonemes and prosody along ventral and dorsal speech streams. *Nature Communications 17(1): 6082*. doi:10.1038/s41467-026-75240-0

2024 **Kim S-G**, De Martino F, Overath T. Linguistic modulation of the neural encoding of phonemes. *Cerebral Cortex 34(4): bhae155*. doi:10.1093/cercor/bhae155

2022 **Kim S-G**[*], Overath T[*], Sedley W, Kumar S, Teki S, Kikuchi Y, Patterson R, Griffiths TD. MEG correlates of temporal regularity relevant to pitch perception in human auditory cortex. *NeuroImage 249: 118879*. doi:10.1016/j.neuroimage.2022.118879 ([*]Both share first authorship)

2020 **Kim S-G**, Poeppel D, Overath T. Modulation change detection in human auditory cortex: Evidence for asymmetric, non-linear edge detection. *European Journal of Neuroscience 52(2): 2889–2904*. doi:10.1111/ejn.14707

2019 **Kim S-G**, Mueller K, Lepsien J, Mildner T, Fritz TH. Brain networks underlying aesthetic appreciation as modulated by interaction of the spectral and temporal organisations of music. *Scientific Reports 9(1): 19446*. doi:10.1038/s41598-019-55781-9

2017 **Kim S-G**, Lepsien J, Fritz TH, Mildner T, Mueller K. Dissonance encoding in human inferior colliculus covaries with individual differences in dislike of dissonant music. *Scientific Reports 7(1): 5726*. doi:10.1038/s41598-017-06105-2

Kim S-G, Knösche TR. On the perceptual subprocess of absolute pitch. *Frontiers in Neuroscience 11: 557*. doi:10.3389/fnins.2017.00557

Kim S-G, Knösche TR. Resting state functional connectivity of the ventral auditory pathway in musicians with absolute pitch. *Human Brain Mapping 38(8): 3899–3916*. doi:10.1002/hbm.23637 (Final draft: https://bit.ly/2I4dbAh)

2016 **Kim S-G**, Knösche TR. Intracortical myelination in musicians with absolute pitch: Quantitative morphometry using 7-T MRI. *Human Brain Mapping 37(10): 3486–3501*. doi:10.1002/hbm.23254

Bianco R, Novembre G, Keller PE, **Kim S-G**, Scharf F, Friederici AD, Villringer A, Sammler D. Neural networks for harmonic structure in music perception and action. *NeuroImage 142: 454–464*. doi:10.1016/j.neuroimage.2016.08.025 (Author's proof: https://goo.gl/ApfnbF)

2011 **Kim S-G**, Kim JS, Chung CK. The effect of conditional probability of chord progression on brain response: An MEG study. *PLoS ONE 6(2): e17337*. doi:10.1371/journal.pone.0017337

*Neuroimaging methods (*N = *9)*

2025 **Kim S-G**. Stimulus-Driven Leakage in Naturalistic Neuroimaging. *Imaging Neuroscience* (accepted). Preprint: doi:10.1101/2025.04.01.646146

2022 **Kim S-G**. On the encoding of natural music in computational models and human brains. *Frontiers in Neuroscience 16: 928841*. doi:10.3389/fnins.2022.928841

2021 Leahy J[*+], **Kim S-G**[*], Wan J, Overath T. An Analytical Framework of Tonal and Rhythmic Hierarchy in Natural Music Using the Multivariate Temporal Response Function. *Frontiers in Neuroscience 15: 665767*. doi:10.3389/fnins.2021.665767 ([*]Both share first authorship; [+]Undergraduate first author)

2014 **Kim S-G**[+], Stelzer J, Bazin P-L, Viehweger A, Knösche T. Group-wise analysis on myelination profiles of cerebral cortex using the second eigenvector of Laplace-Beltrami operator. *Proceedings – IEEE 11th International Symposium on Biomedical Imaging, ISBI 2014, pp. 1007–1010*. doi:10.1109/ISBI.2014.6868043 (Selected for [+]oral presentation)

Chung MK, **Kim S-G**, Schaefer SM, Van Reekum CM, Peschke-Schmitz L, Sutterer MJ, Davidson RJ. Improved statistical power with a sparse shape model in detecting an aging

effect in the hippocampus and amygdala. *Proceedings of SPIE – Progress in Biomedical Optics and Imaging 9034: 90340Y*. doi:10.1117/12.2036497

2012 **Kim S-G**, Lee H, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. Agreement between the white matter connectivity based on the tensor-based morphometry and the volumetric white matter parcellations based on diffusion tensor imaging. *Proceedings - International Symposium on Biomedical Imaging, pp. 42–45*. doi:10.1109/ISBI.2012.6235479 (Acceptance rate: 42%)

Kim S-G, Chung MK[+], Schaefer SM, Van Reekum C, Davidson RJ. Sparse shape representation using the Laplace-Beltrami eigenfunctions and its application to modeling subcortical structures. *Proceedings of the Workshop on Mathematical Methods in Biomedical Image Analysis, pp. 25–32*. doi:10.1109/MMBIA.2012.6164736 (Selected for [+]oral presentation; selection rate: 22%)

2011 **Kim S-G**[+], Chung MK, Seo S, Schaefer SM, Van Reekum CM, Davidson RJ. Heat kernel smoothing via Laplace-Beltrami eigenfunctions and its application to subcortical structure modeling. *Lecture Notes in Computer Science 7087: 36–47*. doi:10.1007/978-3-642-25367-6_4 (Acceptance rate: 42%; selected for [+]oral presentation)

Kim S-G, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. Structural connectivity via the tensor-based morphometry. *Proceedings - International Symposium on Biomedical Imaging, pp. 808–811*. doi:10.1109/ISBI.2011.5872528 (Acceptance rate: 46%)

*Music technology (*N = *2)*

2026 Bjare MR, Bouquillard A, Cantisani G, **Kim S-G**, Lattner S, Widmer G. MEX: The Musical Expectation Benchmark Suite. *Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR), Abu Dhabi, United Arab Emirates* (accepted). https://ismir2026program.ismir.net/poster_293.html

2025 **Kim S-G**, Alonso-Jiménez P, Bechtloff T, Sammler D, Bogdanov D. ManyMusic: An Open-access Music Audio Dataset for Human Experiments on Musical Emotions. *Proceedings – 3rd Workshop on Human-Centric Music Information Research (HCMIR25), Daejeon, Republic of Korea, 20 September 2025*. doi:10.5281/zenodo.17346525

*Translational neuroimaging & clinical applications (*N = *9)*

2021 Weidacker K[*], **Kim S-G**[*], Nord CL, Rua C, Rodgers CT, Voon V. Avoiding monetary loss: A human habenula functional MRI ultra-high field study. *Cortex 142: 62–73*. doi:10.1016/j.cortex.2021.05.013 ([*]Both share first authorship)

Zhang C[*], **Kim S-G**[*], Li J, Zhang Y, Lv Q, Zeljic K, Gong H, Wei H, Liu W, Sun B, Wang Z, Voon V. Anterior limb of the internal capsule tractography: relationship with capsulotomy outcomes in obsessive-compulsive disorder. *Journal of Neurology, Neurosurgery & Psychiatry 92(6): 637–644*. doi:10.1136/jnnp-2020-323062 ([*]Both share first authorship)

2020 Weidacker K[*], **Kim S-G**[*], Buhl-Callesen M, Jensen M, Pedersen MU, Thomsen KR, Voon V. The prediction of resilience to alcohol consumption in youths: insular and subcallosal cingulate myeloarchitecture. *Psychological Medicine, pp. 1–11*. doi:10.1017/s0033291720003852 ([*]Both share first authorship)

2019 Zhang C, **Kim S-G**, Li D, Zhang Y, Li Y, Husch A, Hertel F, Yan F, Voon V, Sun B. Habenula deep brain stimulation for refractory bipolar disorder. *Brain Stimulation 12(5): 1298–1300*. doi:10.1016/j.brs.2019.05.010

Nord CL[*], **Kim S-G**[*], Callesen MB, Kvamme TL, Jensen M, Pedersen MU, Thomsen KR, Voon V. The myeloarchitecture of impulsivity: premature responding in youth is associated with decreased myelination of ventral putamen. *Neuropsychopharmacology 44(7): 1216–1223*. doi:10.1038/s41386-019-0343-6 ([*]Both share first authorship)

Mandali A, Weidacker K, **Kim S-G**, Voon V. The ease and sureness of a decision: evidence accumulation of conflict and uncertainty. *Brain 142(5): 1471–1482*. doi:10.1093/brain/awz013

2017 Yoon YB, Shin W-G, Lee TY, Hur J-W, Cho KIK, Sohn WS, **Kim S-G**, Lee K-H, Kwon JS. Brain structural networks associated with intelligence and visuomotor ability. *Scientific Reports 7(1): 2177*. doi:10.1038/s41598-017-02304-z

2015 **Kim S-G**, Jung WH, Kim SN, Jang JH, Kwon JS. Alterations of gray and white matter networks in patients with obsessive-compulsive disorder: A multimodal fusion analysis of structural MRI and DTI using mCCA+jICA. *PLoS ONE 10(6): e0127118*. doi:10.1371/journal.pone.0127118

2013 **Kim S-G**, Jung WH, Kim SN, Jang JH, Kwon JS. Disparity between dorsal and ventral networks in patients with obsessive-compulsive disorder: Evidence revealed by graph theoretical analysis based on cortical thickness from MRI. *Frontiers in Human Neuroscience*. doi:10.3389/fnhum.2013.00302

*Books (*N = *1)*

2017 **Kim S-G**. *Myeloarchitecture and Intrinsic Functional Connectivity of Auditory Cortex in Musicians with Absolute Pitch.* (Dissertation). In MPI Series in Human Cognitive and Brain Sciences: 184.

*Preprints (*N = *1)*

2026 **Kim S-G**, Müllensiefen D, Yu Y, Overath T. Temporal Structure Perception in Tonal and Atonal Music: An Online Behavioural Study. *PsyArXiv*. doi:10.31234/osf.io/c7vnu

Conference presentations

2026 **Kim S-G**, Overath T, Sammler D. (June). Representational Gradients of Emotion-relevant Musical Information in the Human Cerebral Cortex. *Symposium "Sound and Music: Naturalistic Approaches to Auditory–Motor and Affective Brain Dynamics" (organised by Dr. Sarah Faber), The Annual Meeting of the Organization for Human Brain Mapping (OHBM), Bordeaux, France.*

2025 **Kim S-G**, Alonso-Jiménez P, Bogdanov D, Serra X, Sammler D. (July). Creating an open-access music audio dataset for deep phenotyping of subjective experience of musical emotions. *The 18th International Conference on Music Perception and Cognition (ICMPC), São Paulo, Brazil.*

Kim S-G, Overath T, Sammler D. (March). Representational Gradients of Emotion-relevant Musical Information in the Human Cerebral Cortex. *The 67th Conference of Experimental Psychologists (TeaP), Goethe University Frankfurt, Frankfurt am Main, Germany.*

Kim S-G, Sammler D. (March). How the Sausage is Made: LLM-embedding-based Heuristic Hierarchical Clustering for Organizing TeaP2025 Talk & Poster Sessions. *The 67th Conference of Experimental Psychologists (TeaP), Goethe University Frankfurt, Frankfurt am Main, Germany.*

2024 **Kim S-G**, Overath T, Sammler D. (June). Representational Gradients of Musical Information and Evoked Emotions Revealed by CNN. *The 30th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Seoul, South Korea.*

Kim S-G, Overath T, Sammler D. (June). Representational Gradients of Musical Emotions in the Cerebral Cortex Modeled by a Convolutional Neural Network. *The Neurosciences and Music — VIII, Helsinki, Finland.*

2023 **Kim S-G**, Alonso-Jiménez P, Bogdanov D, Serra X, Sammler D. (October). Exploring Affective Experiences Evoked by Music: Study Plan for a Neurophysiological Deep Dataset "ManyMusic". *CuttingGardens2023-Frankfurt, Frankfurt, Germany.*

Kim S-G, Overath T, Sammler D. (August). Emotion-relevant Representations of Music Extracted by Convolutional Neural Networks Are Encoded in Medial Prefrontal Cortex. *The Joint Conference of the 17th International Conference on Music Perception and Cognition (ICMPC) and the 7th Conference of the Asia-Pacific Society for the Cognitive Sciences of Music (APSCOM), Tokyo, Japan.* (Selected for an oral presentation). doi:10.6084/m9.figshare.24085104

2022 **Kim S-G**, Overath T, Sammler D. (June). Neural Encoding of Musical Emotions Evoked by Naturalistic Stimuli. *The Annual Meeting of the Organization for Human Brain Mapping (OHBM), Glasgow, UK.* doi:10.6084/m9.figshare.20141240.v1

2021 Mu H[+], **Kim S-G**, Overath T. (November). Temporal response function of acoustic energy in fMRI time-series signals. *Advances and Perspectives in Auditory Neuroscience (APAN), Virtual.* ([+]Undergraduate first author)

Yu Y[+], **Kim S-G**, Overath T. (November). Processing temporal structure in music and speech. *Advances and Perspectives in Auditory Neuroscience (APAN), Virtual.* ([+]Undergraduate first author)

Kim S-G, de Martino F, Overath T. (February). Neural encoding of phonemes modulated by linguistic information. *44th Annual MidWinter Meeting, The Association for Research in Otolaryngology (ARO), Virtual.*

2020 **Kim S-G**, de Martino F, Overath T. (October). Neural encoding of phonemes modulated by linguistic information. *Neuromatch Conference, Virtual.* https://youtu.be/6hmtrhIhbOc

Kim S-G, de Martino F, Overath T. (October). Neural encoding of phonemes modulated by linguistic information. *Advances and Perspectives in Auditory Neuroscience (APAN), Virtual.*

2019 **Kim S-G**, Overath T, Sedley W, Kumar S, Teki S, Griffiths TD. (November). MEG correlates of periodicity relevant to pitch perception in human auditory cortex. *The 49th Annual Meeting of Society for Neuroscience (SfN), Chicago, IL, USA.*

Kim S-G, Poeppel D, Overath T. (February). Modulation Change Detection in Human Auditory Cortex: Evidence for Asymmetric, Nonlinear Edge Detection. *The 42nd Annual MidWinter Meeting of the Association for Research in Otolaryngology (ARO), Baltimore, MD, USA.*

2018 **Kim S-G**, Mak EFK, Voon V. (June). Association between microarchitecture and functional topology of cerebral cortex and its behavioral relevance. *The 24th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Singapore.*

2017 **Kim S-G**, Lepsien J, Fritz TH, Mueller K. (June). Inferior colliculus activity correlates with subjective unpleasantness of dissonant music. *The 23rd Annual Meeting of the Organization for Human Brain Mapping (OHBM), Vancouver, Canada.*

2016 **Kim S-G**, Knösche TR. (June). Intrinsic functional connectivity of the ventral auditory pathway correlates with the acuity of absolute pitch. *The 22nd Annual Meeting of the Organization for Human Brain Mapping (OHBM), Geneva, Switzerland.*

Kim S-G, Fritz TH, Lepsien J, Mueller K. (June). Dynamics of functional connectivity in human brains modulated by (un)pleasantness of music. *The 22nd Annual Meeting of the Organization for Human Brain Mapping (OHBM), Geneva, Switzerland.*

2015 **Kim S-G**, Knösche TR. (June). Intracortical myelination in musicians with absolute pitch mapped using 7-T MRI. *The 21st Annual Meeting of the Organization for Human Brain Mapping (OHBM), Honolulu, HI, USA.*

2014 **Kim S-G**, Stelzer J, Schulze K, Viehweger A, Knösche TR. (June). Auditory cortex in musicians with absolute pitch: Deformation-based shape analysis. *The 20th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Hamburg, Germany.*

Kim S-G, Stelzer J, Schulze K, Viehweger A, Knösche TR. (May). Local morphology of auditory cortex in musicians with absolute pitch. *The 5th conference for Neuroscience and Music, Dijon, France.* (Selected for Conference Scholarship)

2013 **Kim S-G**, Stelzer J, Knösche TR. (March). Toward myeloarchitectural analysis on musicians with absolute pitch. *Mind-Brain Symposium, Berlin School of Mind and Brain, Berlin, Germany.*

2012 **Kim S-G**, Chung MK, Jung WH, Jang JW, Kwon JS. (June). Altered properties in the cortical thickness network of patients with obsessive-compulsive disorder. *The 18th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Beijing, China.*

2011 **Kim S-G**, Chung MK, Jung WH, Jang JW, Kwon JS. (November). Application of network analysis based on cortical thickness to obsessive-compulsive disorder patients. *The Biannual Meeting of South Korean Society of Human Brain Mapping (KHBM), Seoul, South Korea.* (Selected for Excellent Oral Award)

Lee H, **Kim S-G**, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (November). Agreement between the white matter connectivity via tensor-based morphometry and the volumetric white matter parcellations. *The 41st Annual Meeting of Society for Neuroscience (SfN), Washington, DC, USA.* (Selected for oral presentation)

Kim S-G, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (June). White matter structural connectivity without diffusion tensor imaging. *The 17th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Quebec, Canada.*

2010 **Kim S-G**, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (November). Structural connectivity via the tensor-based morphometry. *The Biannual Meeting of South Korean Society of Human Brain Mapping (KHBM), Seoul, South Korea.* (Selected for Best Poster Award)

Kim S-G, Kim BS, Kim JS, Chung CK. (March). Differentiating the neural generator of ERAN and MMN: An MEG study. *The 17th International Conference on Biomagnetism, Dubrovnik, Croatia.*

Invited talks

Colloquia & symposia

2024 **Kim S-G**, Overath T, Sammler D. (26 June). Neural encoding of musical emotion. *Symposium, Music Wellness Center, Ewha Womans University, Seoul, South Korea.* (organised by Prof. Eunju Jeong)

2022 **Kim S-G**, Sammler D. (19 July). Investigating the neural encoding of musical emotion using naturalistic stimuli and computational models. *Institutional Colloquium, Max Planck Institute for Empirical Aesthetics, Frankfurt, Germany.* doi:10.6084/m9.figshare.20347794.v1

2021 **Kim S-G**, Sammler D. (9 December). Investigating the neural encoding of musical emotion using naturalistic stimuli. *Brain Imaging Center (BIC) Colloquium (organised by Dr. Ralf Deichmann), Goethe University, Frankfurt, Germany.* (undisclosed)

2019 **Kim S-G**, Michael A, Overath JT. (27 August). Effects of spatially varying residual noise on task-based fMRI GLM. *BIAC (Brain Imaging & Analysis Center) User Meeting (organised by Dr. Allen W Song), Duke University, NC, USA.* (undisclosed)

2016 **Kim S-G**, Knösche TR. (19 September). Auditory cortex in musicians with absolute pitch. *Institutional colloquium, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.* https://bit.ly/2kLbvjh

2015 **Kim S-G**, Knösche TR. (14 September). Relaxometry of absolute pitch. *Mini-symposium: from T1 to myelin case-based discussions on analysis and interpretation (organised by Dr. Robert Trampel), Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.* https://bit.ly/2lWZsPX

2010 **Kim S-G**, Kim JS, Chung CK. (11 September). The effect of conditional probability of chord progression in Western music corpus on brain response: an MEG study. *Joint symposium in celebration of the 20th anniversary of the Korea-Germany Society for Music (organised by Dr. Suk-Won Yi), Seoul National University, Seoul, South Korea.* http://goo.gl/TPGylX

Seminars & workshops

2023 **Kim S-G**. (26 July). Investigating Neural Encoding of Musical Emotions Evoked by Naturalistic Stimuli and Computational Models. *Lab Meeting (Dr. Nathalie Gosselin), International Laboratory for Brain, Music and Sound Research, University of Montreal, Canada.*

2022 **Kim S-G**. (12 August). Investigating Neural Encoding of Musical Emotions Evoked by Naturalistic Stimuli and Computational Models. *Lab Meeting (Dr. Thomas H. Fritz), Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany (virtual).*

2021 **Kim S-G**. (11 May). Structural and functional organizations of pitch perception in human auditory cortex. *Joint Lab Meeting (Dr. Fred Dick and Dr. Maria Chait), University College London, London, UK (virtual).*

2020 **Kim S-G**. (27 July). Predicting the neural encoding of musical structures. *Group Seminar (organised by Dr. Eunju Jeong), Department of Music & Science for Clinical Practice, College of Interdisciplinary Industrial Studies, Hanyang University, Seoul, South Korea.*

Kim S-G. (2 July). Predicting the neural encoding of musical structure. *Group Seminar (organised by Dr. Kyogu Lee), Music and acoustics research group, Graduate School for Convergence Science and Technology, Seoul National University, Suwon, South Korea.* https://bit.ly/3exrQPS

2016 **Kim S-G**. (15 February). In-vivo intracortical myelination mapping: quantitative morphometry. *Group seminar (organised by Dr. Jun Soo Kwon), Seoul National University Hospital, South Korea.* https://bit.ly/3bQGoZn

2015 **Kim S-G**, Knösche TR. (24 August). Intracortical myelination in musicians with absolute pitch. *The Melodic Mind: an interdisciplinary workshop on music and language (organised by Dr. Daniela Sammler), Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.* http://goo.gl/YLcRr8

2011 **Kim S-G**, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (2 December). White matter connectivity without DTI. *Workshop on Diffusion Tensor Imaging and Brain Connectivity (organised by Dr. Moo K Chung), Seoul National University, Seoul, South Korea.* https://bit.ly/2lVngEc

Kim S-G, Chung MK, Schaefer SM, van Reekum C, Davidson RJ. (27 September). Sparse shape representation using the Laplace-Beltrami eigenfunctions and its application to correlating functional signal to subcortical structures. *Workshop on mathematical methods in medical image analysis (organised by Dr. Moo K. Chung), Seoul National University, Seoul, South Korea.* http://goo.gl/sTuutv

Kim S-G, Chung MK, Schaefer SM, van Reekum C, Davidson RJ. (29 July). Correlating EEG to anatomy in hippocampus and amygdala. *Seminar on Brain Network Modeling (organised by Dr. Moo K Chung), Seoul National University Hospital, Seoul, South Korea.* https://bit.ly/2lPimsj

Teaching & mentoring

University courses

2024–2025 "Seminar: Music and Brain", Goethe University Frankfurt, Frankfurt am Main, Germany (co-teaching with Dr. Daniela Sammler)

2022–2024 "Methods in Empirical Musicology", a practical course (seminar and hands-on) for undergraduate and graduate students majoring in Music Informatics, Karlsruhe University of Music (Hochschule für Musik), Karlsruhe, Germany (online)

Teaching initiatives

2026 Casual Teaching Club (CTC), Max Planck Institute for Empirical Aesthetics, Frankfurt am Main, Germany
- Role: Organiser and instructor
- Volunteer instructors and attendees (PhD students, master's students, postdocs), to offer classes and teaching experience that a research institute otherwise lacks
- Monthly sessions, 2 April – 1 October 2026

Lectures & tutorials

2026 **Kim S-G**. (23 September). Workshop: Building a 1,000-Song Naturalistic EEG Dataset: Practical Lessons from ManyMusic. *CuttingGardens 2026, Frankfurt local programme, CoBIC, Frankfurt am Main, Germany.* https://doi.org/10.6084/m9.figshare.34886340

2025 **Kim S-G**. (21 July). Workshop: Linearized Encoding Analysis (LEA)—Predictive Modeling of Music Perception and Emotion in Brain, Body and Behavior. *The 18th International Conference on Music Perception and Cognition (ICMPC), São Paulo, Brazil.* https://s.gwdg.de/yGKPI1

Kim S-G. (9 March). Computing Language for Open Science: Open Discussion. *Open Science Workshop, The 67th Conference of Experimental Psychologists (TeaP), Goethe University Frankfurt, Frankfurt am Main, Germany.* (organised by Kim S-G)

2024 **Kim S-G**. (5 December). Workshop: Music, Brain, and Emotions: An Engaging Self-Experiment. *Max Planck School of Cognition – Cognition Academy, Berlin, Germany.* https://github.com/seunggookim/mpscog-ca25-mbe

Kim S-G. (7 September). Tutorial: Linearized Encoding Analysis. *Korean Society for Music Perception and Cognition (KSMPC) Summer School, South Korea (online).* https://github.com/seunggookim/ksmpc-ss24-sess3

2023 **Kim S-G**. (7 February). Encoding Analysis of fMRI Data. *Data Analysis Lecture Series, Brain Imaging Centre, the University Hospital Frankfurt, Frankfurt am Main, Germany.* https://s.gwdg.de/TJdctg

2021 **Kim S-G**. (27 October). Primer for Linearized Encoding Analysis. *NCML lab meeting, MPIEA, Frankfurt am Main, Germany.* https://s.gwdg.de/wp6LTH

2020 **Kim S-G**. (24 April). Temporal Response Function as Ridge Regression to Estimate Finite Impulse Response. *Overath lab meeting, Duke University, Durham, NC, USA.* https://cli.re/QnJaXw

Kim S-G. (2 February). Introduction to Ridge Regression. *Overath lab meeting, Duke University, Durham, NC, USA.* https://cli.re/5Pa3KZ

2019 **Kim S-G**. (5 December). Robust detrending and inpainting for M/EEG data. *Overath-Woldorff joint lab meeting, Duke University, Durham, NC, USA.* https://bit.ly/2PEd6E5

2017 **Kim S-G**. (8 June). Improving coregistration by skull-stripping of EPI images. *Method Club (organised by Dr. Karsten Mueller), Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.* https://bit.ly/2mimWiW

2015 **Kim S-G**. (15 October). Assessment of “denoising” (motion artifacts removal) on resting state fMRI data. *Method Club (organised by Dr. Karsten Mueller), Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.* https://bit.ly/2kO0eP5

Mentoring

2026 Augustin Bouquillard (intern at NCML, MPI for Empirical Aesthetics, Frankfurt; École Polytechnique, France)
- Co-supervisors: Dr. Daniela Sammler and Dr. Stefan Lattner (Sony CSL Paris)
- Project: MEX: The Musical Expectation Benchmark Suite (ISMIR 2026); behavioural validation of an autoregressive diffusion model estimating entropy and surprisal from music audio
- Duration: 1 April – 31 July 2026

2025–2026 Elana Radtke (student research assistant at NCML, MPI for Empirical Aesthetics, Frankfurt)
- Project: ManyMusic
- Tasks: EEG and fMRI data acquisition

2023 Till Bechtloff (undergraduate intern at NCML, MPI for Empirical Aesthetics, Frankfurt)
- Project: ManyMusic (JQYA funded)
- Tasks: Development of GUI for the online behavioural experiment using jsPsych

2022 Zoë Bolz (undergraduate intern at NCML, MPI for Empirical Aesthetics, Frankfurt)
- Project: Linguistic knowledge and cultural exposure on prosody perception
- Tasks: behavioural data analysis with R

2021–2022 Han (Medy) Mu (undergraduate researcher at O-lab, Duke University, NC)

- Project: Neural encoding of music in fMRI data
- Tasks: MATLAB & Bash for fMRI data processing and linearized encoding analysis

2021–2022 Ying Yu (undergraduate researcher at O-lab, Duke University, NC)
- Project: Temporal structures in tonal and atonal music: online behavioural experiment
- Tasks: MATLAB, JavaScript, HTML for Amazon MTurk experiment data acquisition & analysis

2019–2021 Jasmine Leahy (undergraduate researcher at O-lab, Duke University, NC)
- Project: Neural encoding of musical features in EEG data
- Tasks: MATLAB scripting for EEG experiments and data analysis
- B.Sc. thesis (graduation with distinction): "*Cortical Encoding of Tonal and Rhythmic Hierarchy in Real and Imagined Music Using the Multivariate Temporal Response Function*"
- Current position: M.D. student at the Icahn School of Medicine at Mount Sinai, NY, USA

2018–2020 Jie Wan (graduate researcher at O-lab, Duke University, NC)
- Project: Cortical entrainment to temporal structures in natural speech
- Tasks: MATLAB scripting, EEG data analysis, encoding analysis (temporal response function modelling), statistical tests, data visualisation
- Current position: Ph.D. student at University of California – Irvine, CA, USA

Pedagogical training

2021 Entering Mentoring (March–April 2021, 4 weeks; virtual: Google Classroom)
- Topics: Communication and Feedback; Diversity, Equity, and Inclusion
- Organisers: Dr. Branda Yang, Ms. Eva Gjorgieva (Duke Institute for Brain Sciences, NC, USA)

Service to profession

Editorial board

2025–Present Peer Community in Neuroscience
- Current role: Recommender

2025–Present Peer Community in Psychology
- Current role: Recommender

2025–Present PLOS ONE
- Current role: Academic Editor

2024–Present Music Perception
- Current role: Consulting Editor

2022–2025 Frontiers in Neuroscience, specialty section: Auditory Cognitive Neuroscience
- Role: Review Editor

Guest editor (in alphabetical order; number of handled submissions in parentheses)

2023 PNAS (1)

Ad hoc reviewer for journals (in alphabetical order; number of reviews in parentheses)

2014–Present Addiction Biology (1), Brain and Behavior (2), Brain Connectivity (4), Brain Research (4), Brain Structure and Function (2), Cognitive Affective Behavioral Neuroscience (1), Cognitive Neurodynamics (1), Cortex (1), eLife (1), Frontiers in Aging Neuroscience (3), Frontiers in Computational Neuroscience (1), Frontiers in Neuroanatomy (1), Frontiers in Neuroscience (7), Frontiers in Psychology (5), Heliyon (1), Human Brain Mapping (1), IEEE Transactions on Human-Machine Systems (2), International Journal of Psychophysiology (5), iScience (1), Journal of Affective Disorders (1), Journal of Cognitive Neuroscience (1), Journal of Integrative Neuroscience (3), Journal of Neurophysiology (1), Journal of Neuroscience (2), Journal of New Music Research (1), Journal of Physiological Anthropology (1), Music Perception (2), NeuroImage (2), Neuropsychologia (1), Neuroscience Letters (1), Perception (1), PLOS ONE (4), Psychology of Music (1), Psychophysiology (2), Scientific Data (2), Scientific Reports (4)

Organisation of scientific events

2026 Interdisciplinary workshop "Affective Voice Perception and Synthesis (AVoPS)", MPIEA, Frankfurt am Main, Germany (DFG/NRF bilateral programme, with KAIST, South Korea)

- Role: Organiser

2026 Symposium “Sound and Music: Naturalistic Approaches to Auditory–Motor and Affective Brain Dynamics” (with Dr. Sarah Faber et al.), OHBM 2026, Bordeaux, France
- Role: Co-organiser & Speaker

2024 Open Science Workshop (hybrid), MPIEA, Frankfurt am Main, Germany
- Role: Organiser & Speaker

Member

2023–Present International Conference on Music Perception and Cognition (https://icmpc.org/)
- Abstract reviewer (2023–present), Session chair (2023–present)

2012–Present Organization for Human Brain Mapping (https://www.humanbrainmapping.org/)
- Abstract reviewer (2021–present)

2011–Present Society for Neuroscience (https://www.sfn.org/)

Open source projects

GitHub repo Linearized Encoding Analysis. https://github.com/seunggookim/lea
Cortical surface visualisation. https://github.com/solleo/surfviz
Automatic processing & visualisation using SPM12. https://github.com/solleo/myspm
ANTs parallelisation on Condor. https://github.com/solleo/ANTs-Condor

Referees

Daniela Sammler, Ph.D., Dr. Habil.
Tobias Overath, Ph.D., Prof.
Thomas R. Knösche, Ph.D., Dr. Habil., Prof.

Contact details available upon request.